

Kernel Trick SVM

It is a method that allows the algorithm to handle non-linear data by implicitly mapping it into a higher dimensional space where a linear separation is possible, without the computational cost of explicitly transforming the data.

Common Kernel Function

1) Linear Kernel

$$K(x_i, x_j) = x_i \cdot x_j$$

Use Case: Ideal For the data that is linearly separable

2) Polynomial kernel

$$K(x_i, x_j) = (x_i \cdot x_j + c)^d$$

Use case: Effective for data with curved or polynomial relationships.

3) Radial Basic Function Kernel

(Gaussian kernel)

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2)$$

Use case: The most popular choice for complex, non-linear data with no prior knowledge about its shape, mapping data into an infinite dimensional space

4) Sigmoid Kernel

$$K(x_i, x_j) = \tanh(\alpha(x_i \cdot x_j) + c)$$

Use case: Sometimes used in neural networks due to its properties, but

less common in general SVM applications
due to potential issues.